Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/09/2005
PI Name:	Javanainen, Juha D.Sc.		
Project Title:	Quantum Optics Style Theory for Degenerate Bose and Fermi Gases		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ji@phys.uconn.edu	**Fax:**	FY 860-486-3346
PI Organization Type:	UNIVERSITY	**Phone:**	860-486-3458
Organization Name:	University of Connecticut		
PI Address 1:	Department of Physics, U-3046		
PI Address 2:			
PI Web Page:			
City:	Storrs	**State:**	CT
Zip Code:	06269-3046	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NAG3-2880		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We are currently pursuing two main lines of research. First, we are developing a quantum optics style theory for dynamical phenomena in dilute quantum degenerate Fermi gases in the neighborhood of a Feshabch resonance. Second, we are studying the dynamics of ultralow-temperature gases in optical lattices. Our goal is to generate new theoretical concepts in the physics of degenerate atomic gases, such as a Bose-Einstein condensate or a degenerate Fermi gas realized in alkali metal vapors. In the usual manner of basic theoretical research, when the concepts and ideas arising from this work circulate in the community, new theoretical ideas, experiments, and ultimately new technologies will follow. Prospective applications of interest to NASA include atomic clocks and deep-space navigation.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	
Task Progress:	The project has emerged from the start-up phase. We have initiated two new lines of research, and obtained important results in both of them. (i) We have developed a numerical theory for the conversion of a degenerate Fermi gas of atoms into a gas of molecules during a sweep of the magnetic field across a Feshbach resonance. The key conceptual finding is that the conversion is driven by a dynamical instability of the Fermi sea of atoms against molecule formation. Comparisons with existing experiments are satisfactory, and we have made the prediction that the initial temperature of the atomic gas sets the efficiency limit for the atom-molecule conversion. (ii) We have revisited our old problem of Bose-Einstein condensates in an optical lattice. We have analyzed the “tilted” lattice with a constant force driving the atoms from site to site. In a qualitative agreement with recent experiments, we have found that, surprisingly, the system is unstable in the limit when the driving force is small. We have also continued work that started during the predecessor NASA grant NAG8-1428: (iii) When atoms obeying the Fermi-Dirac statistics are put in an optical lattice, a fractional fermion may result; the system behaves as if it had half of an atom in it. We have shown that such a half-fermion may be detected optically. Finally, during the report period several publications have appeared that originate from incidental and/or discontinued lines of earlier research.
Bibliography Type:	Description: (Last Updated: 01/08/2020)
Articles in Peer-reviewed Journals	Zheng Y, Kostrun M, Javanainen J. "Low-acceleration instability of a Bose-Einstein condensate in an optical lattice." Phys Rev Lett. 2004 Dec 3;93(23):230401. PMID: 15601128 , Dec-2004
Articles in Peer-reviewed Journals	Mackie M, Dannenberg O, Piilo J, Suominen K-A, Javanainen J. "New directions in degenerate dipolar molecules via collective association." European Physical Journal D. 2004 Nov;31(2):273-82. http://dx.doi.org/10.1140/epjd/e2004-00122-3 , Nov-2004
Articles in Peer-reviewed Journals	Ishkhanyan A, Javanainen J, Nakamura H. "A basic two-state model for bosonic field theories with a cubic nonlinearity." Journal of Physics A. 2005 Apr;38(16):3505–16. http://dx.doi.org/10.1088/0305-4470/38/16/003 , Apr-2005
Articles in Peer-reviewed Journals	Javanainen J. "Bose-Einstein condensates interfere and survive." Science. 2005 Mar 25;307(5717):1883-5. PMID: 15790832, , Mar-2005